

TOM MARSH

US Permanent Resident (Green Card holder) - no sponsorship required

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SUMMARY

Senior Machine Learning Engineer with 8 years building production ML systems at TB/PB scale, now working on LLM evaluation and agent-system reliability. Track record delivering measurable business impact: foundational ML infrastructure generating €15M+ annually, platform migrations saving \$200K+ yearly, and 0-1 startup ML system contributing to \$10M+ valuation. Physics background (MSc) with end-to-end ownership from architecture to deployment.

EXPERIENCE

Sprout Social - Senior ML Ops Engineer

Boston, MA | Apr 2026 - Present

- Root-caused unbounded GPU memory growth across a 12-service model-serving fleet to an undocumented ONNX Runtime defect that had shipped for three minor versions: established cold footprint byte-identical across image versions, isolated growth to distinct inference threads rather than concurrent request load, and falsified the concurrency-limiting hypothesis by semaphore tuning; findings drove the remediation and a seven-item SRE/ML action plan
- Built the batch evaluation pipeline at the core of the company's LLM evaluation framework: data model, input loading, parallel executor with deterministic result ordering, and an adaptive AIMD tokens-per-second throttle that backs off on provider rate limits and ramps back up; now the shared path other teams' evaluations run on
- Led CUDA 13 and Python 3.12 migration across 12 GPU inference services, moving both axes simultaneously to close torch/onnxruntime CVEs that were hard-capped by the previous base image, re-resolving the full dependency constraint set across 27 files
- Audited the evaluation framework's datasets against live agent behaviour after tool-schema drift, surfacing three production agent defects (result ordering, chart-type nondeterminism, and relative-date resolution), each reported to the owning squad with trace spans to reproduce
- Redesigned production ML service alerting: root-caused a flapping latency monitor to pod-roll noise rather than client impact, rewrote it as p95 with hysteresis gated on downstream queue backlog, and replaced 12 monitors generating ~90 pages/day with two notify-only ones

Ströer Labs NZ - Senior Machine Learning Engineer

Christchurch, New Zealand | Feb 2018 - Nov 2025

- Reduced ad impression render time by 50% through ML inference optimisation, generating €15M+ additional annual revenue across 3B+ daily ad requests (1T+ annually)
- Led complete migration of ML data pipelines from Spark to DBT, eliminating Spark cluster costs (\$10K+/month), reducing pipeline runtime 85% (20 min to 3 min), and simplifying maintenance for the successor team
- Architected migration from AWS SageMaker to custom Kubernetes infrastructure, enabling GPU training for LightGBM models (previously impossible), unifying ML platform with company infrastructure
- Designed and deployed real-time bid optimization system selecting demand-side partners or dropping traffic entirely, operating under p99 sub-20ms latency at 100K+ requests/second
- Cut cloud egress costs by 35% (\$200K+ annually) for product vertical through ML optimisation; maintenance team expanded savings to \$30K/month post-transition
- Led 0-1 development of multiple ML products (classification, time-series forecasting, optimisation) from concept to production on PB-scale datasets, establishing technical architecture before transitioning to dedicated teams
- Built and owned ML experimentation platform through multiple iterations (custom webapp, Grafana dashboards, MLFlow), enabling stratified A/B performance analysis across 3B+ daily ad impressions and driving data-informed model iteration

- Built model monitoring dashboards with stratified performance views, SHAP-based feature importance reporting, and compliance documentation to support measurable, explainable, and auditable ML decisions
- Drove ML platform modernisation during the company's bare-metal-to-AWS migration, collaborating with DevOps and Data Engineering to redesign datalakes, account structures, and deployment practices
- Designed technical interview process for ML Engineering roles: created programming assessments, architecture evaluation framework, and systems design questions used across MLE/SWE/DevOps hiring
- Conducted 80+ technical interviews over 6 years, contributing to 15 hires as the NZ office grew from 20 to 30 engineers
- Hosted interns each summer and mentored juniors through intermediate

Pyper Vision - Founding Machine Learning Engineer (Contracts, part-time)

Contract 1 | Christchurch, New Zealand | Jul 2024 - Feb 2025

- Established complete ML infrastructure as sole technical hire, enabling aerospace startup pivot from hardware to ML-based fog forecasting (company expanded 5 to 20+ airports, valued \$10M+ post-funding)
- Architected end-to-end system on GCP from scratch: automated data collection, preprocessing pipelines, model training infrastructure, and production inference API with <100ms p99 latency
- Evaluated and deployed model architectures (LightGBM GBDT, TensorFlow CNNs), achieving commercially competitive forecast accuracy, delivering a production system in 8 months
- System scaled to UK market expansion (advised on integration strategy post-contract)

Contract 2 | Christchurch, New Zealand | Oct 2025 - Nov 2025

- Designed a physics-validated fog classification system using a pseudo-labelling pipeline derived from meteorological research papers
- Implemented clustering analysis by airport fog patterns and model error modes, creating a systematic debugging framework for the operations team to diagnose regional forecast failures

EDUCATION

MSc, Physics - University of Canterbury | Jul 2022 - Jul 2024

- Thesis: Extreme precipitation events under climate change using 4000+ years of simulation data at multiple resolutions
- Collaboration with NZ's National Institute of Water and Atmospheric Research (NIWA)
- Publication: Harrington, Rosier, **Marsh**, Frame (2024), "[Robust changes to the wettest and driest days of the year are hidden within annual rainfall projections: a New Zealand case study](#)", *Environmental Research Letters*, 19, 074057
- Presented research at the Meteorological Society of New Zealand Annual Conference (2023)

BSc, Physics & Computer Science - University of Canterbury | Feb 2015 - Nov 2017

- Capstone: Y_2SiO_5 thin film growth for quantum information devices using Pulsed Laser Deposition

TECHNICAL SKILLS

ML: A/B Testing, Experimentation Platforms, SHAP/Explainability, Statistics, Data Mining, Deep Learning (PyTorch, TensorFlow), Gradient Boosting (LightGBM, XGBoost), Classical ML stack (Scikit-learn, Pandas, NumPy), MLFlow

LLM & Agent Systems: LLM Evaluation (LLM-as-judge, offline/online evals), Agent Systems (MCP), OpenTelemetry GenAI conventions

Infrastructure: AWS (SageMaker, EMR, S3, Glue, Athena, Bedrock), GCP (Cloud Functions, GCS), Docker, Kubernetes, CUDA/GPU serving, Datadog, Source Control (Git), Linux, MacOS

Data Engineering: Apache Airflow, Kafka, SQL (Postgres, Presto, Trino), time-series databases, Spark, DBT

Languages: Python, SQL, Java, Scala, Kotlin, Go
